

Medical Equipment Utilization & Maintenance Risk Monitoring System for Kenyan County Hospitals

Predictive maintenance and usage accountability for critical medical equipment

1. Team Members and Roles

Name	Role	Responsibility
Brenda Chepkirui	Solutions Engineer / Data Analyst	Problem definition, data strategy, analytics workflow, business translation
Valentine Nyankwoga	Data Engineer	Data pipeline, ingestion, feature engineering, database design
Caleb Munyekenye	ML Engineer	Risk-prediction model development and evaluation
Member 4-to be updated	Dashboard / Frontend Engineer	Facility risk dashboard and alerting interface

2. Problem Statement

In Kenya's public county hospital system, the problem of undetected equipment breakdown risk, prolonged idle time, and unreliable usage reporting for critical medical equipment — dialysis machines, ICU equipment, imaging and radiotherapy machines — leads to interrupted life-sustaining care, wasted public investment, and weakened financial accountability over a multi-billion-shilling national equipment program.

This is a documented, current problem, not a hypothetical one. A 2026 national audit of Kenya's Managed Equipment Services (MES) dialysis program, which examined hospital records and machine usage data from 2015 to 2021, found that in almost eight out of ten hospitals, the machines' own usage counters showed far more work performed than what was recorded in the hospitals' paper registers — in some cases by more than 30 percent. Separately, machines across 125 hospitals under the same Sh38–63 billion scheme ground to a halt due to lack of

maintenance, prompting government termination of the contracts, while multi-billion-shilling equipment including CT scanners and mammogram machines sit idle in several counties due to staffing or infrastructure gaps. The problem is current: in January 2025 a dialysis machine breakdown at Naivasha sub-county hospital forced patients to seek care elsewhere, and in 2025 Kenyatta National Hospital's main cancer radiotherapy machine failed unexpectedly, disrupting oncology services nationally.

The gap is a monitoring and accountability problem, not an absence of underlying data.

Hospitals already log equipment usage, whether through machine counters or paper registers, and the Ministry of Health's Kenya Health Facility Census already records facility-level equipment inventory and service availability nationally. What is missing is a systematic way to compare recorded usage against expected or counter-verified usage, flag facilities where equipment sits idle or is at elevated breakdown risk, and shift maintenance from reactive (fix after failure) to condition-based (intervene before failure) — exactly the kind of gap the 2026 MES audit exposed after the fact, at national scale.

Currently, equipment monitoring relies on manual paper registers and reactive maintenance triggered only after a breakdown has already disrupted patient care. This approach fails because paper-based reporting is demonstrably unreliable at scale, as the counter-versus-register discrepancy shows, and because reactive maintenance means a machine is only inspected once it has already stopped working — by which point patients have already been turned away or delayed.

Proposed approach: use data engineering and machine learning to combine facility-level equipment inventory data, usage/maintenance logs, and documented incident patterns into a risk-monitoring system that flags equipment at elevated breakdown risk and highlights usage-reporting discrepancies, aiming to shift facilities from reactive to condition-based maintenance and improve accountability over public equipment investment.

3. Domain & Context

Industry: Healthcare — public hospital medical equipment asset management

Specific Operational Context: Kenya's county hospital network operating under the national Managed Equipment Services (MES) program and its successor, the National Equipment Support to Counties (NESC) scheme — covering dialysis, ICU, radiology, and theatre equipment leased or supplied to public hospitals across all 47 counties. This is critical care infrastructure: dialysis and ICU equipment in particular are life-sustaining, and an unplanned outage directly and immediately affects patient survival, not just service quality.

The operational challenge sits at the intersection of asset integrity monitoring (predicting breakdown before it happens) and financial/usage accountability (reconciling actual usage against what is reported) — both surfaced by the same 2026 national audit, and both addressable from the same underlying usage and maintenance data, which is what makes a single MVP viable across both angles.

4. Data Source

Real, facility-level equipment usage and maintenance logs are not bulk-published by Kenya's Ministry of Health or individual county hospitals. The data strategy therefore separates real, publicly available context and method-validation data from a small, clearly-labeled set of incident-grounded scenarios used for evaluation.

Layer 1 — Real data, for context and method validation

- Kenya Health Facility Census (Ministry of Health): a real, national government dataset recording facility-level equipment inventory, service availability, and infrastructure gaps — used to build a realistic facility and equipment master list and to derive features such as equipment type, county, and reported service availability.
- A public equipment-degradation telemetry dataset used purely to validate the breakdown-risk prediction method — rolling usage/condition features leading up to failure — on real sensor degradation data, independent of anything the team constructs. This is a method check, not a healthcare-specific claim.

Layer 2 — Incident-grounded scenario data, for the demo

Simulated facility-level equipment usage and maintenance logs constructed to reproduce two specific, documented patterns from the Kenyan case history:

- A usage-versus-reporting discrepancy pattern, modeled on the 2026 MES audit finding of counter-vs-register mismatches
- A maintenance-neglect-to-breakdown pattern, modeled on the Naivasha and MES 125-hospital equipment-halt cases

Each scenario is generated with a written rationale tracing it to a specific cited case, so the dataset is auditable rather than arbitrary.

What this data strategy explicitly does not claim: the project will not present its risk scores as validated against real Kenyan hospital equipment outcomes at scale, since real facility usage/maintenance logs were not accessible. It will claim the underlying method is validated on real degradation data (Layer 1) and demonstrated against realistic, cited Kenyan scenarios (Layer 2) — a distinction stated explicitly in the final presentation, consistent with the standard applied to every other option considered for this capstone.

5. Success Metrics

Technical metrics (model performance)

Metric	Target	Measured against
Breakdown-risk detection rate on labeled scenario cases	> 90%	Scenario dataset (Layer 2)
Lead time before predicted failure window	Flag risk at least 5–10 usage-days before simulated breakdown	Scenario dataset (Layer 2)
Usage-reporting discrepancy detection rate	> 85% of injected discrepancy cases flagged	Scenario dataset (Layer 2)

Lead time before failure is the headline metric, not accuracy alone — it is the number that maps directly to the business case: under reactive maintenance, a facility only learns a machine has failed once it has already stopped working, so any demonstrated lead time is the core result worth defending.

Business metrics (impact framing)

Area	Expected impact if deployed
Patient safety	Fewer unplanned outages of life-sustaining equipment (dialysis, ICU), reducing forced patient transfers and treatment delays
Financial accountability	Usage-reporting discrepancies flagged systematically, supporting oversight of the KSh38–63 billion national equipment investment
Equity of access	Idle or underused equipment surfaced for reallocation, addressing the documented pattern of multi-billion-shilling equipment sitting unused in some counties
Maintenance efficiency	Shift from reactive (post-failure) to condition-based (pre-failure) maintenance scheduling

As with the technical metrics, these are framed as potential impact given the MVP is built on public census data and a labeled scenario set, not a live hospital deployment — the presentation should state this plainly.

6. Initial Timeline (Weeks 7–12)

Week	Focus	Deliverable
7	Problem framing & data acquisition — finalize MVP scope to one equipment type (dialysis, given the richest documented case history); acquire Kenya Health Facility Census data and the C-MAPSS degradation dataset; design scenario logic tied to cited cases	Data architecture document + initial dataset extracts
8	Data engineering pipeline — clean and join facility census data, build the usage/maintenance log schema, ingest C-MAPSS data	Working ingestion pipeline producing a unified facility-equipment dataset
9	Feature engineering & scenario dataset build — usage trends, maintenance interval features, reporting-discrepancy features; finalize labeled scenario set	Feature-engineered dataset (Layer 1 + Layer 2) ready for modeling
10	Model development — train and tune breakdown-risk model and discrepancy-flagging logic, evaluate against target metrics	Trained risk model with documented evaluation results
11	Dashboard & alerting — facility/equipment risk view, discrepancy flags, lead-time-to-failure display, wired to model output	Working risk dashboard demoing prioritized facility/equipment risk
12	Final testing & presentation — end-to-end validation, business narrative tied to the 2026 MES audit and documented breakdown cases, rehearsal	Final capstone presentation and demo

This timeline reflects a single, focused MVP scoped to one equipment type (dialysis machines) with one combined risk-and-discrepancy capability — sized realistically for a four-person team across six weeks. Extending to additional equipment types (ICU, imaging, radiotherapy) is explicitly named as future work, not attempted this term.